

Growth without more CO_2?

Which countries achieved absolute decoupling between 2000 and 2022?

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Date: October 13, 2026

Introduction

A country's economic growth is strongly linked with a higher energy demand and therefore more carbon dioxide (CO_2) emissions. In order to measure the economic growth of a country in combination with the decrease in resource consumption that it creates and its environmental impact, the term widely used is "decoupling" [1]. To measure economic growth alone, we measure the country's rise in GDP.

If economic growth and gas emissions are strongly correlated, there is a conflict between a country's climate targets, and economic goals are in direct conflict. For a country to achieve **absolute decoupling**, its GDP must increase consistently while CO_2 emissions fall [1]. In this report, I am addressing the question: **Which countries achieved absolute between 2000 and 2022, and how common it is?**

Data sample

The dataset [2] is the *CO_2 and Greenhouse Gas Emissions* dataset from Our World in Data. It contains 50 411 rows and 79 columns, with one row per country and year, covering 1750–2024. The only columns that were used for the results include the country name, the ISO code, the year, the GDP and the CO_2 emissions (territorial, in million tonnes.

For the cleaning of the dataset, aggregates like "World" or "Europe" were removed. The years examined included the year 2000 and 2022 and the countries that were kept included both GDP and CO_2 values in the chosen years. The rest of the countries were taken into consideration. After the performance of the cleaning, the countries left were just 164.

Analysis method. For each country I calculated the percentage change in GDP and in CO_2 emissions between 2000 and 2022. Each country was classified as having achieved either *absolute decoupling* (GDP up and CO_2 down), or *relative decoupling* (both up, but CO_2 slower), or *no decoupling* (rest). In order to test the correlation between the two values, Spearman rank correlation was used and the analysis was performed with the R language.

Results and discussion

Table 1 and Figure 1 summarize the results. Of the 164 countries, 42 (25.6%) achieved absolute decoupling. The statistical uncertainty, considering the Poisson distribution, is $\sqrt{42} \approx 6.5$ countries. The median raised its GDP by 63.9% while lowering its emissions by 21.9%. 80 countries (48.4%) were only able to reach a relative decoupling, with the rest 42 countries (25.6%) not able to achieve any decoupling, since the median emissions was raised by 244%.

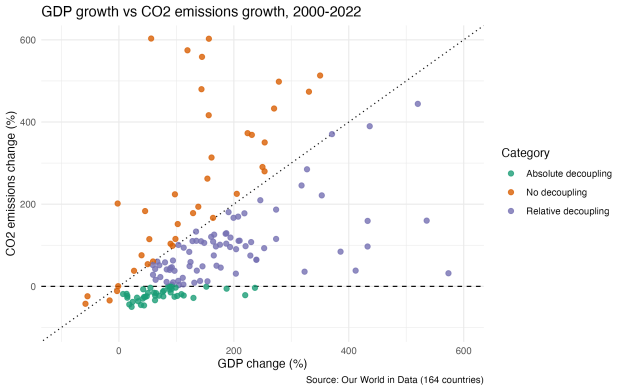


Figure 1: Change in GDP versus change in CO_2 emissions between 2000 and 2022 for 164 countries. Points below the dashed line reduced their emissions; points below the dotted line have CO_2 growing more slowly than GDP. A few extreme countries lie outside the plotted range.

Category	N	Share (%)	Median ΔGDP (%)	Median ΔCO_2 (%)
Absolute decoupling	42	25.6	+63.9	−21.9
Relative decoupling	80	48.8	+167	+87.4
No decoupling	42	25.6	+141	+244

Table 1: Number of countries per category, 2000–2022.

GDP growth and CO_2 growth are strongly and positively correlated, as shown from the Spearman coefficient $\rho = 0.67$, $p < 10^{-15}$. The main outcome of this analysis is that economic growth is strongly liked, after all, with more emissions and that the absolute decoupling is the exception, since only one in four countries is able to achieve it. The strongest reductions, as shown in Figure 2, are mostly found in Europe; Denmark (−46.4% CO_2 with +43.7% GDP) and the United Kingdom (−45.3% with +38.4%). Ukraine ($_4$9.9%) is not a characteristic example, since this percent is mainly due to war and economic collapse.

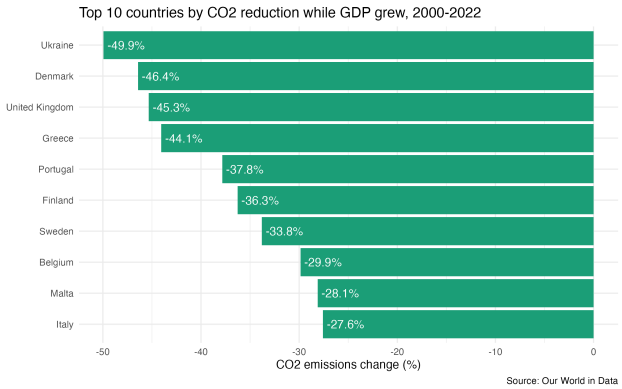


Figure 2: The ten largest reductions in CO_2 emissions among the countries with absolute decoupling, 2000–2022.

Two limitations should be noted, including that I am comparing only 2 years and the changes in between are ignored, and that the emissions are accounted only for each country's territory, so a country can be placed higher if it moves the production abroad.

References

[1] J. Mossé and C. Ramos. *Decoupling and green growth*. Carbone 4, September 2021. Available at: https://carbone4.com/en/publications/publication-decoupling (accessed October 2026).

[2] H. Ritchie, P. Rosado and M. Roser. *CO_2 and Greenhouse Gas Emmissions*. Our World in Data. Available at: https://github.com/owid/co2-data. (accessed October 2026).